## Task 1: Network Design Simulation – Operation Process and Test Results

### 1. Network Topology Design

In Task 1, a small routed network topology was designed using Cisco Packet Tracer. The topology consists of two routers, two switches, and four PCs. The network is divided into three IP subnets:

- Left LAN: 192.168.1.0/24
- Inter-router network: 192.168.2.0/24
- Right LAN: 192.168.3.0/24

PC0 and PC1 are connected to Switch0, which is connected to Router1. PC2 and PC3 are connected to Switch1, which is connected to Router2. Router1 and Router2 are connected through an intermediate Layer 2 switch. This switch only forwards frames at the data link layer and does not participate in routing, so logically the two routers are considered to be directly connected.

### 2. IP Address Configuration

Each device was configured with an IP address according to the task requirements.

**Router1**

- GigabitEthernet0/0: 192.168.1.1 /24
- GigabitEthernet0/1: 192.168.2.1 /24

**Router2**

- GigabitEthernet0/0: 192.168.2.2 /24
- GigabitEthernet0/1: 192.168.3.1 /24

**PC Configuration**

- PC0: 192.168.1.10 /24, Gateway 192.168.1.1
- PC1: 192.168.1.11 /24, Gateway 192.168.1.1
- PC2: 192.168.3.10 /24, Gateway 192.168.3.1
- PC3: 192.168.3.11 /24, Gateway 192.168.3.1

### 3. Router Configuration

The routers were configured using the Cisco IOS command line interface. Each interface was assigned the correct IP address and enabled using the no shutdown command.

Static routing was then configured to allow communication between the two LANs.

On Router1, a static route was added to reach the 192.168.3.0/24 network via Router2:

- Destination: 192.168.3.0/24
- Next hop: 192.168.2.2

On Router2, a static route was added to reach the 192.168.1.0/24 network via Router1:

- Destination: 192.168.1.0/24
- Next hop: 192.168.2.1

After configuration, the routing tables and interface status were checked using the show ip interface brief command to ensure all required interfaces were in the up/up state.

### 4. Connectivity Verification

After completing the configuration, network connectivity was verified using the ping command.

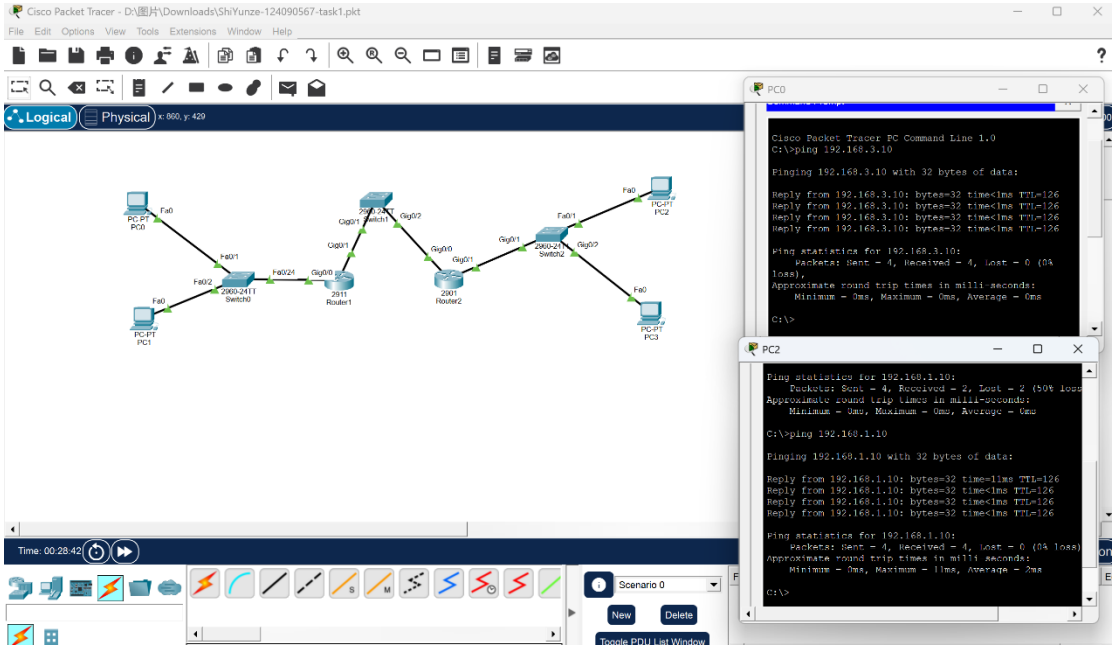
- PC0 was able to successfully ping PC1 within the same subnet.
- PC2 was able to successfully ping PC3 within the same subnet.
- PC0 was able to successfully ping PC2 and PC3 across different subnets.

- PC2 was able to successfully ping PC0 and PC1 across different subnets.

These results confirm that routing between the two LANs was correctly implemented.

**5. Conclusion**

In this task, a two-router network topology was successfully designed and implemented. By correctly configuring IP addresses, default gateways, and static routing entries, full connectivity between different network segments was achieved. The experiment demonstrates the basic principles of routed network design and verifies the role of routers in enabling communication between different IP subnets.




**Task 2: VLAN Network Simulation – Operation Process and Test Results**

**1. Network Topology Design**

In Task 2, a VLAN-based network topology was designed using Cisco Packet Tracer. The topology consists of three Layer 2 switches and six PCs. No router or Layer 3 switch was used in this task. The purpose of this experiment is to demonstrate logical network segmentation using VLANs.

The devices are organized as follows:

- Switch0 hosts PCs in VLAN 10
- Switch1 hosts PCs in VLAN 20
- Switch2 hosts PCs in VLAN 30

Each PC is connected to its corresponding switch using copper straight-through cables.

**2. VLAN Creation and Port Assignment**

VLANs were created on each switch using the Cisco IOS command line interface. Each VLAN represents a separate broadcast domain.

The following VLANs were configured:

- VLAN 10 on Switch0
- VLAN 20 on Switch1
- VLAN 30 on Switch2

After creating the VLANs, switch ports connected to PCs were configured as access ports and assigned to the appropriate VLAN.

This ensures that each PC belongs to the correct VLAN and that broadcast traffic is isolated within the same VLAN.

**3. IP Address Configuration**

Each PC was manually assigned an IP address according to its VLAN membership. PCs in the same VLAN were configured within the same IP subnet.

- VLAN 10:
  - PC0: 192.168.10.10 /24
  - PC1: 192.168.10.11 /24
- VLAN 20:
  - PC2: 192.168.20.10 /24
- VLAN 30:
  - PC3: 192.168.30.10 /24
  - PC4: 192.168.30.11 /24
  - PC5: 192.168.30.12 /24

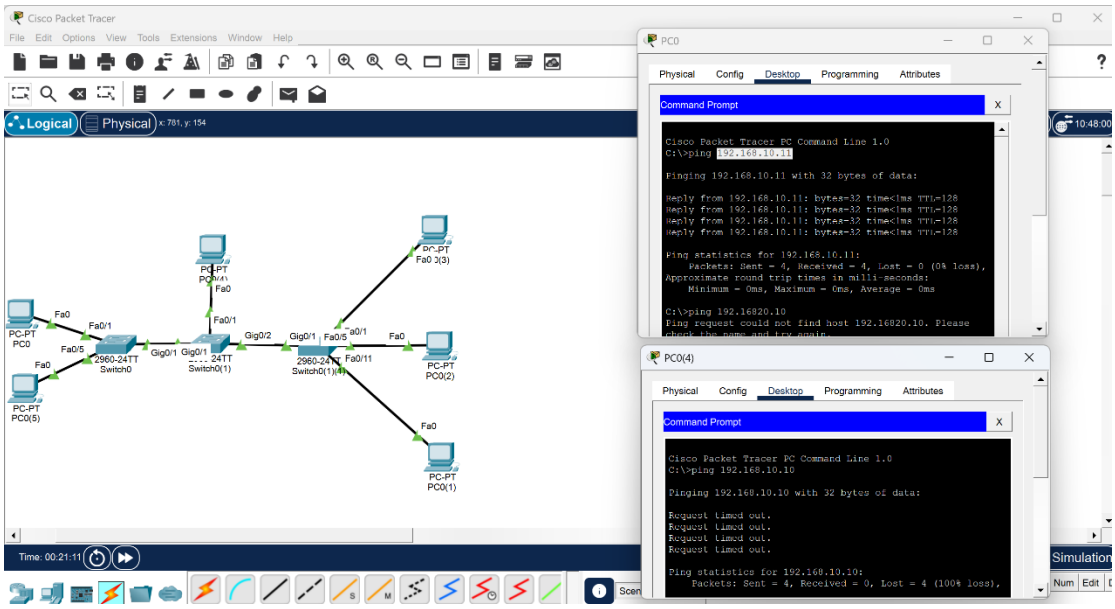
No default gateway was configured because there is no Layer 3 device to perform inter-VLAN routing.

**4. Connectivity Verification**

After completing the VLAN and IP configuration, connectivity was tested using the ping command.

- PCs within the same VLAN were able to communicate successfully.
- PCs in different VLANs were unable to communicate with each other.

For example, PC0 could successfully ping PC1 in VLAN 10, while PC0 could not ping PC2 in VLAN 20. This behavior is expected because VLANs separate broadcast domains and no inter-VLAN routing was configured.

**5. Conclusion**

This task demonstrates how VLANs can be used to logically segment a network on Layer 2 devices. By creating VLANs and assigning switch access ports accordingly, devices in different VLANs are isolated from each other. The experiment confirms that inter-VLAN communication requires a router or a Layer 3 switch, highlighting the importance of VLANs in network design and management.